CareRoute: local AI intake and staff handoff

Delivered version: local AI prototype, policy demo-policy-0.3. **Record date:** 25 October 2026. Invented records only.

What is delivered

CareRoute reads an invented appointment request with a real language model running on the laptop, then gives staff a structured assessment and exact source excerpts. Staff can inspect the original input, record a review, confirm a synthetic appointment and inspect the persisted audit history.

The original build defaulted to no AI and had never executed a live provider call. This version defaults to local inference, provides a model-readiness panel and displays the actual model, duration and output-token count. Scenario buttons fill editable inputs; they do not replay stored assessments. Authored replay remains explicitly labelled in advanced controls.

Model and architecture

The default model is **qwen3:4b-instruct**, a roughly 2.5 GB Q4_K_M model served by Ollama 0.40.11. Real inference was verified on the NVIDIA RTX 4060 Laptop GPU through Vulkan, with the loaded model using 3.61 GiB of GPU memory at an 8,192-token context. Runtime binaries and weights are installed separately and excluded from the source archive.

The browser sends an invented request to the Python standard-library server. SQLite records receipt first. The server asks the local model for JSON conforming to an enum schema, validates the complete response and exact evidence offsets, applies versioned rules, and persists the result. The staff workspace presents the result beside the original description.

The six model assessments are current warning, appointment intent, information sufficiency, contradiction, supported demonstration scope and human-review indication. The handoff is extractive: the model selects up to three exact excerpts rather than producing medical advice. Fabricated quotes, extra fields, invalid labels and incomplete generations are rejected.

Local inference is serialized, has a 90-second deadline and a 512-token output limit. Only loopback Ollama endpoints and installed local GGUF models are accepted. Local mode requires no API key and makes no hosted-model call. Requests remain saved for staff review if the model is busy, missing, offline or invalid.

Completed workflow

1. Load an invented scenario or enter invented information. Edit the text freely.
2. Submit and receive a durable receipt while the actual AI assessment runs.
3. Inspect model-selected excerpts, provenance and the administrative suggestion.
4. Sign in as the demo reviewer, inspect the original request and record a human review.
5. Select a future time and confirm a synthetic booking. Inspect the audit history.

Immediate warning checks run before validation, persistence or model completion. The model cannot suppress a structured emergency indication. Staff review and appointment booking remain separate actions. Late AI cannot overwrite a human takeover, and an emergency downgrade requires a different supervisor.

Executed engineering checks

Check	Recorded result	Meaning
Automated policy, API, storage and adapter suite	134/134 passed	Includes actual HTTP/SQLite workflows and mocked failure contracts
Browser regression checks	12/12 passed	Form, emergency checks offline, replay honesty, review, booking, audit, labels and mobile
Live-model browser checks	5 passed	Real model response through UI, persisted provenance, review and booking, outage fallback, mobile
Actual local-model evaluation	43/43 valid responses	Fresh inference, raw responses and failures retained
Additional semantic probes	8/8 passed	New wording, negation, history, vague input, intent, scope, warning paraphrase and embedded instructions
Route agreement on comparable authored references	36/38	Visible development examples; not clinical accuracy

The successful local inference calls took **1.50 seconds median and 1.64 seconds p95** in this run. These measurements include local HTTP handling and generation and reuse an already-loaded model. Cold startup can take longer; these are not latency guarantees. The live browser run separately exercises persistence and human workflow. All records are newly invented.

Paired comparison with no AI

Metric	No-AI baseline	Local AI with shared rules
Emergency-route misses	0/13	0/13
Incorrect routine suggestions / nonroutine references	0/32	1/32
Escalations / routine references	6/6	0/6
Execution failures	0	0

Comparable route disagreements: **FX-14, FX-28**. Five fixtures with AI-fault/output-specific references are excluded from input-disposition comparisons, with their reasons retained in the raw evaluation. They remain in execution totals. FX-31 represents failed initial persistence and is tested in the API/browser suite instead of this model runner.

This evaluation uses visible, authored development examples. It does not establish clinical safety, a held-out accuracy score, calibration or measured staff-time savings. Source research and the pre-local-AI report remain available as historical material; they are not substituted for this implementation's evidence.

The warning-paraphrase probe intentionally falls outside the immediate phrase matcher. If recognized by the model, its warning appears only after inference. This is an explicit coverage/timing limit; the model does not make pre-submit detection universal. Semantic probe failures: none in this run.

Probe pass criteria cover the final route, selected warning/intent fields and exact excerpts, not every intermediate judgment. The unsupported-scope probe still had an incorrect additional-review flag. The warning-paraphrase probe missed the contradiction with all-no form answers. Raw outputs are retained, and shared rules apply independently of the model's auxiliary flags.

The first prompt passed 5/8 semantic probes. The prompt was refined using these visible examples, and the current run passed 8/8. Earlier attempts remain archived. Both the original cases and these extra probes informed development; neither set is held out.

Reproduce and run

On this machine, double-click Start CareRoute.cmd. On a fresh Windows checkout, run the setup script, then the launcher:

```
powershell -ExecutionPolicy Bypass -File tools/install_local_ai.ps1
python -X utf8 tools/start_demo.py
```

Open http://127.0.0.1:8080. Demo staff credentials are reviewer / demo-reviewer; supervisor / demo-supervisor is for second review. Python 3.11 or later is required. No third-party Python package is needed to run the app. Browser checks and PDF rendering use Playwright with installed Chrome/Edge or Chromium.

```
python -X utf8 tools/run_checks.py
python -X utf8 tools/browser_check.py
python -X utf8 tools/browser_live_check.py
python -X utf8 tools/evaluate_live.py --execute --arms local_ai
python -X utf8 tools/probe_local_ai.py --execute
python -X utf8 tools/summarize_local_run.py
python -X utf8 tools/render_report.py
python -X utf8 tools/build_submission.py
```

The submission ZIP includes source, scripts, reports and a verified SHA-256 manifest. It excludes databases, credentials, caches, downloaded model weights and runtime binaries. The original research and specification snapshots are preserved.

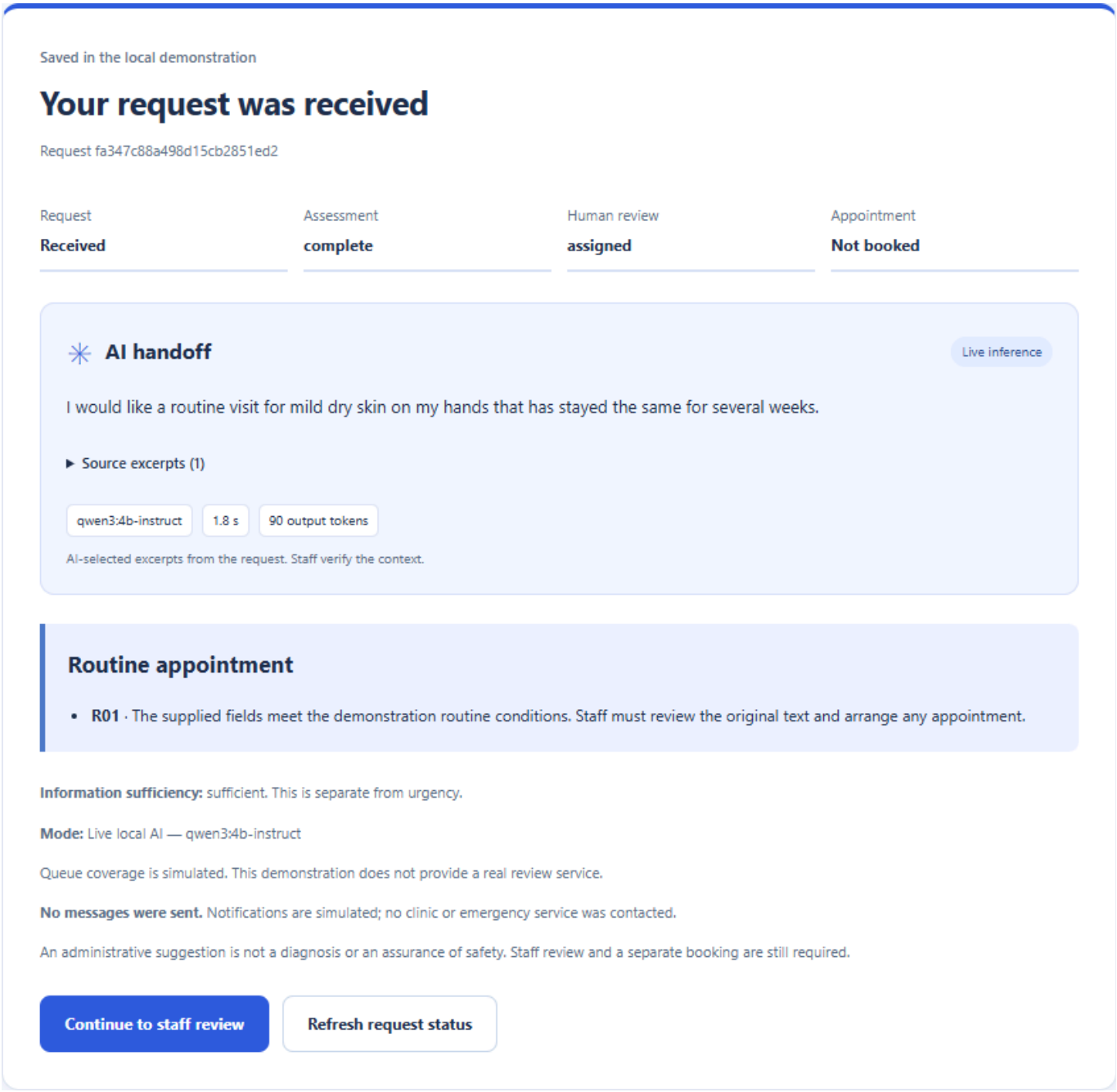
Limits and sources

CareRoute is a local hackathon demonstration. It is not a medical service and must use invented records. Its narrow example scope and emergency phrases are illustrative, not approved clinical policy. Real authentication, clinical validation, deployment operations, notifications and health-system integration are outside this submission. All appointments and coverage are simulated.

- Recorded test results and browser regression results.
- Live browser evidence and local-model summary.
- Semantic probes and runtime/GPU evidence.
- Actual model run with every input identifier, raw response, decision and failure.
- Ollama model card and structured chat API.
- Original report before local AI and demo walkthrough.

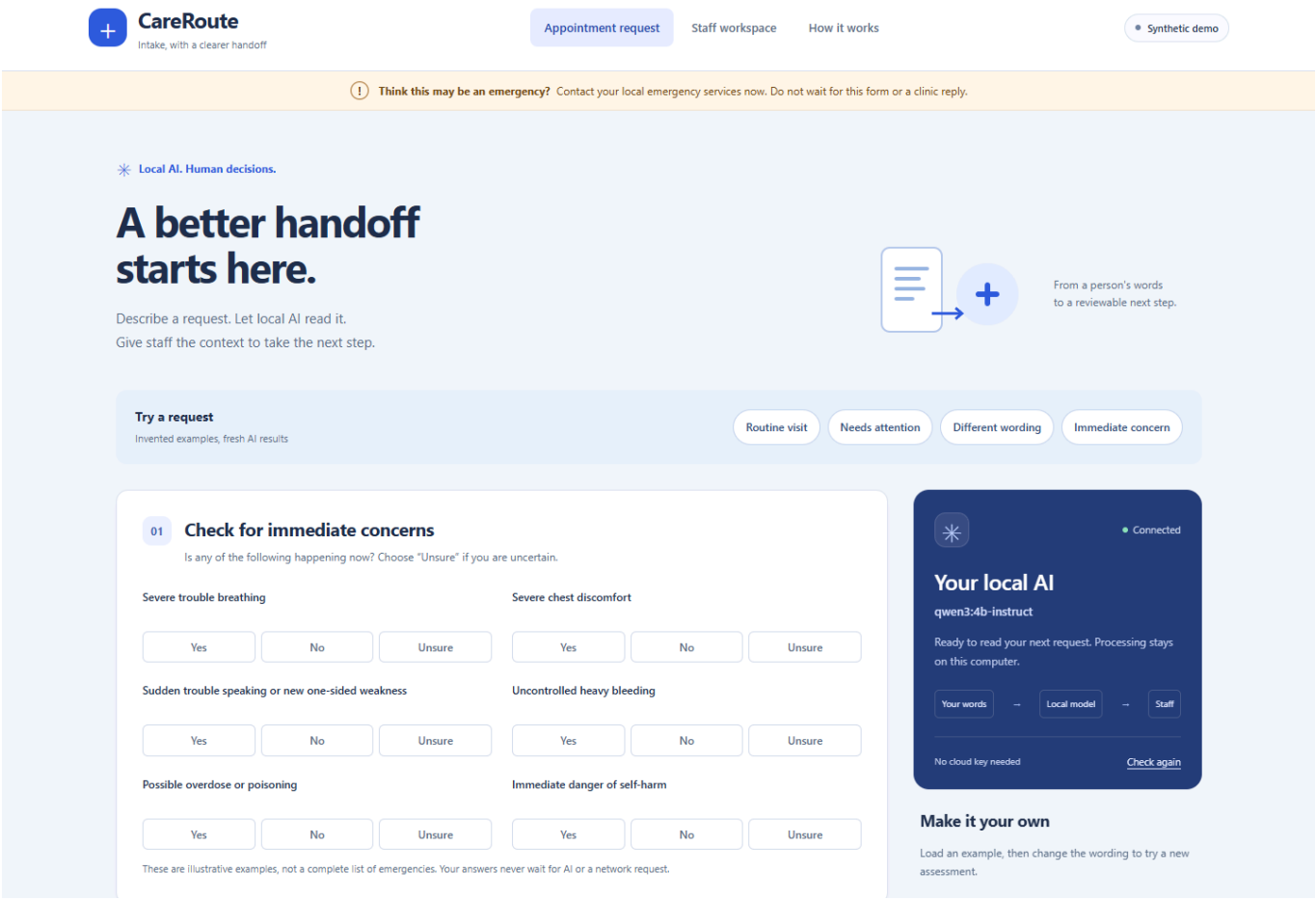
Actual local AI result

Actual screenshot of the synthetic prototype; no patient records.



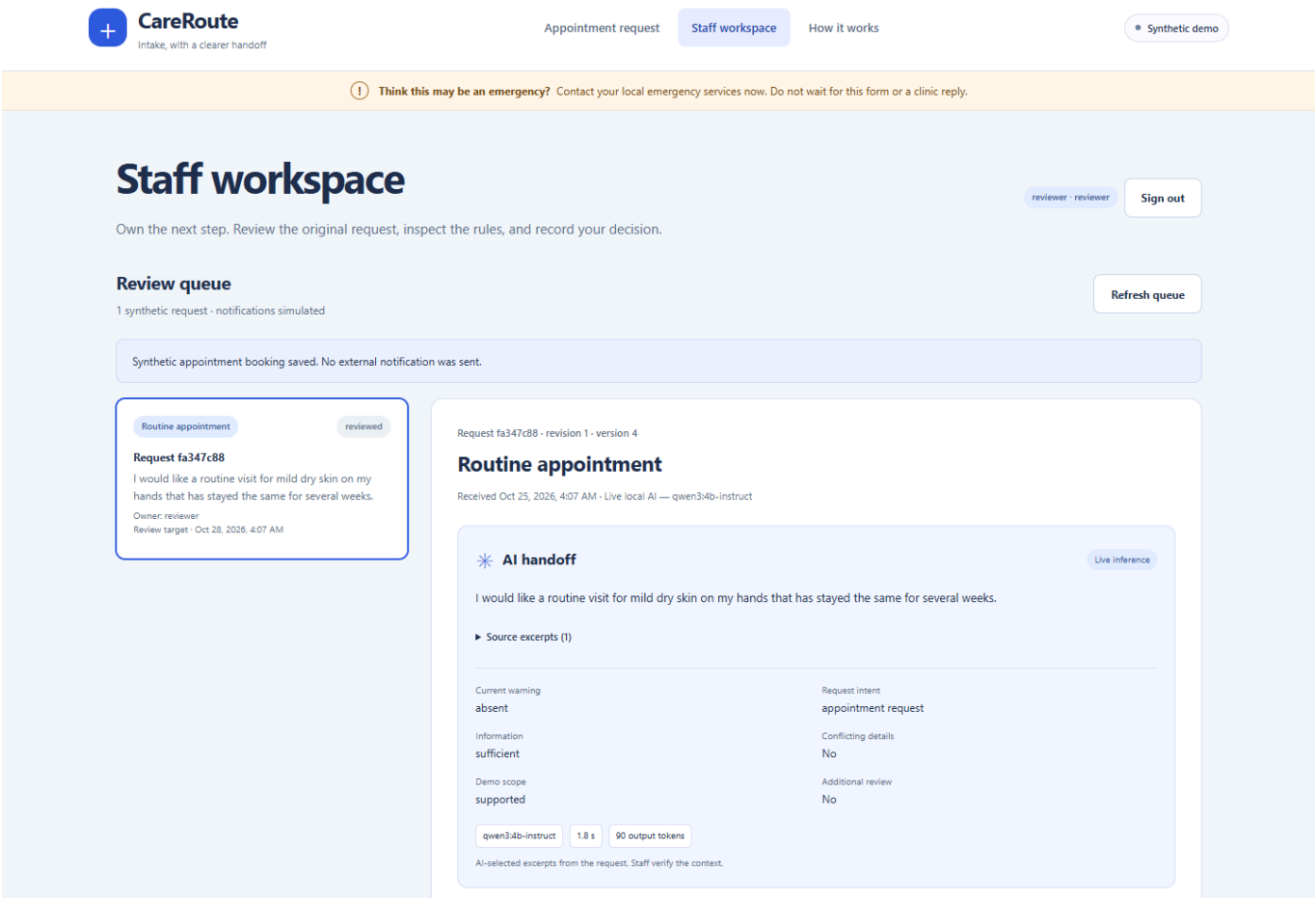
Desktop intake and model status

Actual screenshot of the synthetic prototype; no patient records.



Staff review and AI handoff

Actual screenshot of the synthetic prototype; no patient records.



Mobile intake

Actual screenshot of the synthetic prototype; no patient records.

